

Physical modelling of Yb:YAG spectroscopic Mueller-matrix data

Methods, conventions, validation and pitfalls

Analysis pipeline `fit/` — Yb:YAG project, FTMC

August 3, 2026

Abstract

This document describes, in full, how the refractive index, the Yb^{3+} absorption spectrum and the scattering loss of two Yb:YAG slabs were extracted from J. A. Woollam RC2 Mueller-matrix data. The analysis replaces an earlier wavelength-by-wavelength inversion with a single Kramers–Kronig-consistent oscillator model fitted globally to ten angles of incidence and the normal-incidence transmittance simultaneously. Emphasis is placed on the parts that are easy to get wrong: the incoherent treatment of the backside beam, the Kramers–Kronig partner of the Gaussian oscillator, the fact that the instrument’s own error bars are unusable as fitting weights, the separation of genuine absorption from instrumental structure, and the decision to fix every Yb line centre at its predicted Stark-level position rather than fit it. Each substantive claim is accompanied by the numerical test that supports it.

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1 Scope and summary

1.1 What was measured

Two uncoated Yb:YAG slabs, labelled A and B, were measured on a J. A. Woollam RC2 rotating-compensator ellipsometer over 210–1700 nm:

- **Reflection:** the full normalised Mueller matrix at ten angles of incidence, 30° – 75° in 5° steps (sample B).
- **Transmission:** normal-incidence intensity transmittance (samples A and B).

1.2 What is extracted

1. $n(\lambda)$, the refractive index, from the angular dependence of the reflection Mueller matrix. This is independent of the slab thickness.
2. The per-pass optical depth αd , split into a Kramers–Kronig-consistent *absorption* part carried by Yb^{3+} oscillators and a non-KK *scattering* part.
3. The Yb^{3+} oscillator strengths, with line centres fixed at values predicted from the crystal-field level scheme.

1.3 Why a global model rather than point-by-point inversion

The previous analysis let n and k float independently at each of ~ 1000 wavelengths. Two free parameters per wavelength can always reproduce the data, so the fit was necessarily good — but it could not answer the question it raised. The modelled transmittance ran 2–3% high in the near infrared, meaning the loss required to explain T exceeded what k could carry without spoiling the reflection fit. With n, k free at every point there is nothing to prevent absorption and scattering trading against each other, and nothing to identify which is which.

Constraining the whole spectrum to one model closes the trade:

- the ten-angle reflection Mueller matrix fixes $n(\lambda)$ absolutely, making no reference to the thickness;
- the transmittance fixes the *total* per-pass loss $(\alpha_{\text{abs}} + \alpha_{\text{sc}})d$;
- Kramers–Kronig consistency forbids the absorptive part from taking an arbitrary spectral shape — any k rising towards the NIR must appear in n — so the smooth remainder that cannot be accommodated is pinned as scattering.

2 Measurements and file handling

2.1 The export format

The CompleteEASE .dat exports are whitespace-separated, with one repeating block per (wavelength, angle):

E	λ	θ	Ψ	Δ	σ_Ψ	σ_Δ			
dPolE	λ	θ	depol		σ				
mm11	λ	θ	value		σ				(raw intensity, R or T)
mm12 ... mm44	λ	θ	value	value	σ	σ			(normalised to mm11)

A parsing trap. The files carry a UTF-8 byte-order mark followed by CRLF line endings, so the first physical line contains only the BOM and the `VASEmethod[...]` header sits on the *second* line. Any parser that inspects only the first line silently fails to find the header. The reader opens the files with encoding `utf-8-sig` and scans all lines for the header token.

2.2 Reflection or transmission?

CompleteEASE writes `TransmissionMeas=1` into the header for transmission scans and omits the key entirely for reflection. **The filename is not reliable:** `YbYag_5a_MMt` and `YbYag_5a_UMMt` are named with a trailing “t” but carry no transmission flag and were acquired at 30° – 75° , i.e. they are reflection scans. The header, not the name, decides.

2.3 Data screening

Every file was screened before use. The results are recorded in the code (`SAMPLES` and `UNUSABLE` in `run_fit.py`) so they are not rediscovered later.

Sample	Status	Evidence
B	full $R + T$ fit	reflection at 10 AOI, clean; transmittance clean
A	T only	no reflection scan; transmittance clean
5 %	unusable	transmission files contain <code>inf</code> , negative and > 1 values (the entire 2026-06-23 session returned all NaN); <i>both</i> 5a reflection scans read $m_{12} \approx 0$, $m_{33} \approx -1$ at 30° – 50° , i.e. $\Psi = 45^\circ$, $\Delta = 180^\circ$, no polarisation change — a failed alignment
10 %	unusable	both transmission files contain negative and > 1 transmittance; no reflection scan exists

A further file, `YbYag_B_UMMt` [2026-06-16,175112], has median $T = 0.835$, sitting at the lossless-slab limit; it appears to be a straight-through baseline rather than a sample scan and is excluded.

Both A and B are Yb-doped. Both show the 940 nm pump band at $\alpha d \approx 0.54$ against a ≈ 0.36 baseline. “Pure” in the project notes means *uncoated*, not undoped.

A physicality filter is applied to the transmittance: a slab cannot transmit more than the lossless Fresnel limit $(1 - R_F)/(1 + R_F)$ nor less than zero, and points outside are discarded as failed acquisitions.

3 Optical forward model

3.1 Geometry and conventions

The sample is a plane-parallel slab of complex index $N_1 = n + ik$ in air ($N_0 = 1$), of thickness d in the millimetre range. The time convention is $e^{-i\omega t}$, so a forward-propagating wave carries $\exp(i 2\pi N \cos \theta z / \lambda)$ and $\text{Im } N > 0$ attenuates.

The refraction angle inside the slab follows from Snell’s law with a complex index,

$$\sin \theta_1 = \frac{N_0 \sin \theta_0}{N_1}, \quad \cos \theta_1 = \sqrt{1 - \sin^2 \theta_1}, \quad (1)$$

where the square-root branch is chosen so that $\text{Im } \cos \theta_1 \geq 0$; this keeps evanescent and absorbed waves decaying rather than growing.

3.2 Fresnel coefficients

For an interface $a \rightarrow b$,

$$r_s = \frac{N_a c_a - N_b c_b}{N_a c_a + N_b c_b}, \quad r_p = \frac{N_b c_a - N_a c_b}{N_b c_a + N_a c_b}, \quad (2)$$

$$t_s = \frac{2N_a c_a}{N_a c_a + N_b c_b}, \quad t_p = \frac{2N_a c_a}{N_b c_a + N_a c_b}, \quad (3)$$

with $c_a = \cos \theta_a$, $c_b = \cos \theta_b$.

3.3 From Jones to Mueller

An isotropic sample has a diagonal Jones matrix $\text{diag}(a_p, a_s)$. Its (unnormalised) Mueller matrix is

$$\mathbf{M} = \begin{pmatrix} \frac{1}{2}(I_p + I_s) & \frac{1}{2}(I_p - I_s) & 0 & 0 \\ \frac{1}{2}(I_p - I_s) & \frac{1}{2}(I_p + I_s) & 0 & 0 \\ 0 & 0 & \text{Re } c & \text{Im } c \\ 0 & 0 & -\text{Im } c & \text{Re } c \end{pmatrix}, \quad (4)$$

with $I_p = |a_p|^2$, $I_s = |a_s|^2$ and $c = a_p a_s^*$. Written in ellipsometric angles this is the familiar block-diagonal form

$$\mathbf{M}/M_{11} = \begin{pmatrix} 1 & -\cos 2\Psi & 0 & 0 \\ -\cos 2\Psi & 1 & 0 & 0 \\ 0 & 0 & \sin 2\Psi \cos \Delta & \sin 2\Psi \sin \Delta \\ 0 & 0 & -\sin 2\Psi \sin \Delta & \sin 2\Psi \cos \Delta \end{pmatrix}. \quad (5)$$

Equation (5) is the origin of the symmetry relations used in Sec. 6.1: $m_{12} = m_{21}$, $m_{33} = m_{44}$, $m_{34} = -m_{43}$.

3.4 Incoherent superposition of backside beams

The slab is millimetre-thick and fringe-free at the RC2's spectral resolution. The front-surface beam and the beams that have made one or more round trips inside the slab therefore add **incoherently**: one sums Mueller matrices, not Jones amplitudes. This is the same physics as CompleteEASE's "model backside reflections" switch, and it is what produces both the measured depolarisation and the angular fanning of the pseudo-index. Neglecting it makes the inferred n too low.

Writing r, t for the front face (ambient $\rightarrow$ substrate), r', t' for the same face from inside, and r_b for the back face, the m -th emergent amplitude is

$$a_j^{(m)} = t_j t'_j r_{b,j} (r_{b,j} r'_j)^{m-1} \beta^{2m}, \quad j \in \{s, p\}, \quad (6)$$

where the single-pass amplitude factor is

$$\beta = \exp\left(\frac{i 2\pi N_1 \cos \theta_1 d}{\lambda}\right) \exp\left(-\frac{1}{2} \frac{\alpha_{\text{sc}} d}{\cos \theta_1}\right). \quad (7)$$

The first exponential carries both the optical phase and the absorption implied by $\text{Im } N_1$; the second carries the non-KK scattering loss introduced in Sec. 4.5 (an intensity loss, hence the

factor $\frac{1}{2}$ in the amplitude). The total normalised reflection Mueller matrix is

$$\mathbf{M}_{\text{tot}} = \left[\mathbf{M}(r_p, r_s) + \sum_{m=1}^{N_b} \mathbf{M}(a_p^{(m)}, a_s^{(m)}) \right] / M_{\text{tot},11}, \quad (8)$$

with $\mathbf{M}(\cdot, \cdot)$ given by Eq. (4). $N_b = 3$ round trips are retained by default; the series converges rapidly because $|r_b r'| \lesssim 0.01$ and $|\beta|^2 \approx 0.7$.

Because $\mathbf{M}$ depends only on $|a|^2$ and $a_p a_s^*$, the optical phase in Eq. (7) cancels identically in the sum — which is precisely the statement that the beams are incoherent.

3.5 Transmittance

For transmission the geometric series can be summed in closed form. Per polarisation,

$$T_j = \frac{\mathcal{T}_{f,j} \mathcal{T}_{b,j} A}{1 - \mathcal{R}_{i,j} \mathcal{R}_{b,j} A^2}, \quad A = \exp\left(-\frac{4\pi \text{Im}(N_1) d}{\lambda \cos \theta_1}\right) \exp\left(-\frac{\alpha_{\text{sc}} d}{\cos \theta_1}\right), \quad (9)$$

where $\mathcal{T}$ and $\mathcal{R}$ are *intensity* coefficients. The interface transmittances carry the flux factor $\mathcal{T} = |t|^2 \text{Re}(N_b c_b) / \text{Re}(N_a c_a)$. The reported transmittance is the unpolarised average $\frac{1}{2}(T_s + T_p)$, which at normal incidence is exact by symmetry.

3.6 Optional surface roughness

A nanometre-scale roughness layer, if enabled, is treated *coherently* (it is far thinner than the coherence length) as a Bruggeman effective medium of 50% void in the substrate material,

$$f_1 \frac{\varepsilon_1 - \varepsilon_e}{\varepsilon_1 + 2\varepsilon_e} + f_2 \frac{\varepsilon_2 - \varepsilon_e}{\varepsilon_2 + 2\varepsilon_e} = 0, \quad (10)$$

solved on the branch with $\text{Im} \varepsilon_e \geq 0$. The three-phase ambient/roughness/substrate stack gives effective front coefficients

$$r_{012} = \frac{r_{01} + r_{12} e^{2i\beta r}}{1 + r_{01} r_{12} e^{2i\beta r}}, \quad t_{012} = \frac{t_{01} t_{12} e^{i\beta r}}{1 + r_{01} r_{12} e^{2i\beta r}}, \quad (11)$$

which then enter the incoherent slab sum in place of the bare Fresnel coefficients. Setting $d_{\text{rough}} = 0$ reduces these identically to the bare interface. In the final fits roughness is held at zero: n already matches the single-crystal reference to 0.0018 without it, so there is nothing for it to absorb.

3.7 Validation of the forward model

`python fit/slab.py` runs four independent checks:

Test	Expected	Measured
Lossless slab, normal incidence	$T + R = 1$	exact to 10^{-6} printed
Isotropy of the reflection MM	$m_{12} - m_{21} = 0$	0 (identically)
	$m_{33} - m_{44} = 0$	0
	$m_{34} + m_{43} = 0$	0
	$m_{12} = -\cos 2\Psi$	4.4×10^{-16}
Opaque limit $\rightarrow$ single interface	$m_{33} = \sin 2\Psi \cos \Delta$	2.2×10^{-16}
$d_{\text{rough}} \rightarrow 0$ continuity	bare interface	4.9×10^{-15}

4 Dielectric-function model

4.1 Requirement: Kramers–Kronig consistency

Causality links the real and imaginary parts of ε ,

$$\varepsilon_1(E) - \varepsilon_\infty = \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{E' \varepsilon_2(E')}{E'^2 - E^2} dE'. \quad (12)$$

Enforcing Eq. (12) is what makes the absorption/scattering separation possible: an absorptive feature cannot be added to k without a corresponding, measurable signature in n .

The model is therefore built entirely from KK-consistent components:

$$\varepsilon(E) = \varepsilon_\infty + \sum_p \frac{A_p}{E_{c,p}^2 - E^2} + \sum_g \mathcal{G}(E; A_g, E_g, \text{Br}_g). \quad (13)$$

4.2 Sellmeier poles and the reference seed

A pole below the measured range contributes $\varepsilon_1 = A/(E_c^2 - E^2)$ with no absorption. Two are used, one ultraviolet and one infrared, seeded from the Zelmon single-crystal YAG dispersion [1],

$$n^2 = 1 + \frac{2.28200 \lambda^2}{\lambda^2 - 0.01185} + \frac{3.27644 \lambda^2}{\lambda^2 - 282.734}, \quad [\lambda] = \mu\text{m}. \quad (14)$$

Conversion to energy space uses

$$\frac{B\lambda^2}{\lambda^2 - C} = \frac{BE_c^2}{E_c^2 - E^2}, \quad E_c = \frac{hc}{\sqrt{C}}, \quad (15)$$

giving $E_c = 11.388 \text{ eV}$, $A = 296.0 \text{ eV}^2$ for the UV pole and $E_c = 0.07373 \text{ eV}$, $A = 0.01781 \text{ eV}^2$ for the IR pole. The seed reproduces the published n to 5×10^{-6} at all test wavelengths, which verifies the conversion algebra.

4.3 The Gaussian oscillator and its KK partner

The Yb^{3+} $4f$ – $4f$ lines are inhomogeneously broadened, so a Gaussian lineshape is appropriate. Its KK partner is analytic in terms of the Dawson function $D(x) = e^{-x^2} \int_0^x e^{t^2} dt$. With $\sigma = \text{Br}/(2\sqrt{\ln 2})$,

$$\varepsilon_2(E) = A \left[e^{-\left(\frac{E-E_g}{\sigma}\right)^2} - e^{-\left(\frac{E+E_g}{\sigma}\right)^2} \right], \quad (16)$$

$$\varepsilon_1(E) = \frac{2A}{\sqrt{\pi}} \left[D\left(\frac{E+E_g}{\sigma}\right) - D\left(\frac{E-E_g}{\sigma}\right) \right]. \quad (17)$$

The second, negative-energy term in Eq. (16) enforces the odd symmetry $\varepsilon_2(-E) = -\varepsilon_2(E)$ that Eq. (12) demands; omitting it biases ε_1 at low photon energy.

Why not a Lorentzian. A Lorentzian has $\varepsilon_2 \sim E^{-2}$ wings that would spread absorption from the 940 nm band across the entire transparent window, where the measurement shows none. The Gaussian's e^{-x^2} wings vanish far faster and keep the Yb absorption local.

4.4 Numerical verification of KK consistency

Equations (16)–(17) are verified numerically by transforming ε_2 with Eq. (12) on a 4×10^5 -point grid and comparing to the analytic ε_1 . The principal value is evaluated by singularity subtraction, which is exact and remains well behaved *on* resonance where a naively masked grid does not:

$$\mathcal{P} \int_0^L \frac{xf(x)}{x^2 - a^2} dx = \int_0^L \frac{xf(x) - af(a)}{x^2 - a^2} dx + \frac{f(a)}{2} \ln \frac{L - a}{L + a}, \quad (18)$$

using $\mathcal{P} \int_0^L dx/(x^2 - a^2) = (2a)^{-1} \ln[(L - a)/(L + a)]$. Agreement is $\sim 10^{-10}$ off resonance and 2.4×10^{-5} at the line centre, the latter limited by grid discretisation. Run with `python fit/dispersion.py`.

4.5 Scattering: a deliberately non-KK channel

A ceramic or defect haze removes light from the specular beam by *redirecting* it, not by absorbing it. It is therefore not KK-consistent, and forcing it into k distorts both n and the oscillator strengths. It is carried separately as an intensity loss with a **fitted exponent**,

$$\alpha_{\text{sc}}(\lambda) = C \left(\frac{1000 \text{ nm}}{\lambda} \right)^p. \quad (19)$$

The exponent is fitted rather than fixed at the Rayleigh value $p = 4$, and this matters. The measured loss falls only $\approx 20\%$ from 300 to 1650 nm, whereas $p = 4$ would vary by a factor ~ 900 across that span. A fixed λ^{-4} term is consequently driven to zero by the fit and leaves the NIR transmittance $\approx 2\%$ under-predicted — exactly the discrepancy the earlier point-by-point analysis reported. The fitted p is thus a diagnostic of scatterer size, not a nuisance knob.

5 Yb³⁺ spectroscopy: predicting the line positions

5.1 Stark manifolds

Yb³⁺ has a single excited multiplet. In YAG the crystal field splits the two manifolds into [2]

$$^2F_{7/2} \text{ (ground) : } 0, 565, 612, 785 \text{ cm}^{-1}, \quad (20)$$

$$^2F_{5/2} \text{ (excited) : } 10327, 10634, 10927 \text{ cm}^{-1}. \quad (21)$$

Every allowed electronic absorption line is a difference of one level from each set, so its **position is predicted, not fitted**. At room temperature the 565, 612 and 785 cm⁻¹ levels are thermally populated, producing hot bands to the red of the zero-phonon line (ZPL, $0 \rightarrow 10327$, 968.3 nm).

Boltzmann populations at 300 K ($kT = 208 \text{ cm}^{-1}$) relative to the ground level are 0.066, 0.053 and 0.023 for the three excited ground-state levels, so the hot bands are expected to be weak and the 785 cm⁻¹ ones weakest.

5.2 Vibronic sidebands

Phonon-assisted transitions add replicas displaced to the blue of the parent electronic line by one phonon energy. YAG optical phonons span roughly 150–800 cm⁻¹. Three replicas are included, and they are *declared as vibronic*, which is what entitles them — and only them — to be broad:

Replica	Displacement	Centre
ZPL +150 cm ⁻¹	lowest YAG optical phonon	954.5 nm
ZPL +420 cm ⁻¹		930.5 nm
10927 + 500 cm ⁻¹	wing of the 915 nm line	875.1 nm
10927 + 200 cm ⁻¹		898.7 nm

5.3 Merging unresolved transitions

Transitions closer than 2 nm are merged onto a single intensity-weighted centre, because the band is not resolved below that separation. Kept separate they are mutually degenerate: the fit cannot apportion strength between them and convergence stalls without any gain in information. The threshold must sit below the 3.2 nm spacing of the 965/968/969 nm group, which the instrument *does* resolve — merging that group leaves +0.026 residual on the sharp ZPL and −0.011 on its blue neighbour.

6 The inverse problem

6.1 Observables and isotropy symmetrisation

From Eq. (5), an isotropic sample must satisfy $m_{12} = m_{21}$, $m_{33} = m_{44}$ and $m_{34} = -m_{43}$. The fit therefore uses the symmetric combinations, which carry all the physical signal:

$$s_{12} = \frac{1}{2}(m_{12} + m_{21}), \quad s_{33} = \frac{1}{2}(m_{33} + m_{44}), \quad s_{34} = \frac{1}{2}(m_{34} - m_{43}), \quad (22)$$

and reserves the antisymmetric combinations

$$a_{12} = \frac{1}{2}(m_{12} - m_{21}), \quad \text{etc.} \quad (23)$$

for uncertainty estimation, since the model predicts them to be identically zero.

6.2 Realistic uncertainties

The instrument’s reported error bars cannot be used as fitting weights. The RC2 reports $\sigma \sim 10^{-4}$ on the normalised Mueller elements and $\sim 6 \times 10^{-4}$ on transmittance. The true systematic is roughly seventy times larger, and two independent diagnostics prove it:

1. $m_{12} - m_{21}$ is consistently −0.0075 across all files, although isotropy requires the difference to vanish;
2. $\approx 10\%$ of measured m_{12} values fall *below* −1, which a normalised Mueller matrix cannot physically do.

The error assigned to each symmetric observable is therefore

$$\sigma_s = \sqrt{\sigma_{\text{reported}}^2 + \langle a^2 \rangle}, \quad (24)$$

i.e. the reported value added in quadrature with the RMS of the corresponding antisymmetric combination — a direct, model-free measurement of the instrument systematic. Transmittance receives a 0.5% photometric floor (`--t-sys`).

Consequence. Left at face value the transmittance block outweighs the entire ten-angle reflection data set by a factor ~ 100 , and n is sacrificed to chase a systematic: the fit then returns $n(1550 \text{ nm}) = 1.73$. With honest floors it returns 1.8081, against the single-crystal reference 1.8070.

6.3 Block balancing

Reflection contributes $\sim 3 \times 10^4$ residuals and transmittance $\sim 10^3$. Each block is scaled by $1/\sqrt{N_{\text{block}}}$ so that the two data types contribute comparably regardless of how many points each happens to contain; a user-settable multiplier `--w-trans` allows this to be overridden.

6.4 Cost function

The fit minimises

$$\chi^2 = \frac{1}{N_R} \sum_{\text{blocks } b} \sum_{\lambda, \theta} \left(\frac{s_b^{\text{mod}} - s_b^{\text{meas}}}{\sigma_{s_b}} \right)^2 + \frac{w_T}{N_T} \sum_{\lambda} \left(\frac{T^{\text{mod}} - T^{\text{meas}}}{\sigma_T} \right)^2. \quad (25)$$

6.5 Parameterisation, bounds and the fixed centres

The free parameters are the two pole amplitudes and the UV pole energy, the scattering pair (C, p) , and for each oscillator its amplitude and width.

Line centres are fixed, not fitted. This is the single most important choice in the analysis. When centres were free, an oscillator drifted onto whatever residual bump was nearby and parked itself at 955 nm with a 52 nm width against its bound — a “crystal-field line” broader than the entire manifold structure it was supposed to describe. Fixing the centres also makes the problem tractable: with free centres neighbouring lines are mutually degenerate and the optimiser needed 697 function evaluations instead of 45.

Widths are bounded by *kind*: electronic lines ≤ 0.025 eV, vibronic sidebands ≤ 0.100 eV. Without this split the fit again parks a very broad “line” against its bound and uses it to absorb whatever the line list is missing.

6.6 Optimiser

`scipy.optimize.least_squares`, trust-region reflective, with `x_scale='jac'` to handle the $\sim 10^{-5}$ amplitudes and $\sim 10^2$ pole amplitudes in one vector. The forward model is fully vectorised over the $[\lambda, \theta]$ grid, so one residual evaluation costs a few tens of milliseconds and the whole fit converges in 45 evaluations.

6.7 Detection limit and the prune-and-refit pass

An oscillator whose amplitude refines to zero has a width with *zero gradient*. A single such parameter makes the entire Jacobian singular and returns NaN for *every* uncertainty in the fit. The pipeline therefore performs a second pass: lines with $A < 10^{-7}$ (corresponding to $\alpha d \sim 3 \times 10^{-4}$, more than an order of magnitude below the photometric floor) are dropped and the fit repeated. They are reported as “not detected” rather than silently retained.

6.8 Parameter uncertainties

One-sigma values come from the Jacobian at the solution,

$$\text{cov} = (\mathbf{J}^\top \mathbf{J})^{-1} \frac{2 \text{cost}}{N_{\text{res}} - N_{\text{par}}}, \quad (26)$$

scaled by the residual variance because the weights are relative rather than absolute standard deviations.

7 The thickness degeneracy

A fringe-free thick slab measures only the *product* αd . The thickness is an input (`--thickness`, default 1 mm), and:

- $n(\lambda)$ is **independent** of d ;
- k , all oscillator amplitudes and both scattering coefficients scale as $1/d$.

The `alphad_abs` and `alphad_scatter` columns of the output CSV are the thickness-independent quantities and are the ones to quote. Once d is measured with a micrometer, multiply through once.

8 Results

8.1 Refractive index

n matches the Zelmon single-crystal YAG dispersion to **RMS** 0.0018 (maximum deviation 0.0047) over 350–1650 nm — from a global model, not a per-wavelength inversion. Fitted dispersion parameters: UV pole $A = 256.0$ eV² at $E_c = 10.640$ eV; IR pole $A = 0.0018$ eV².

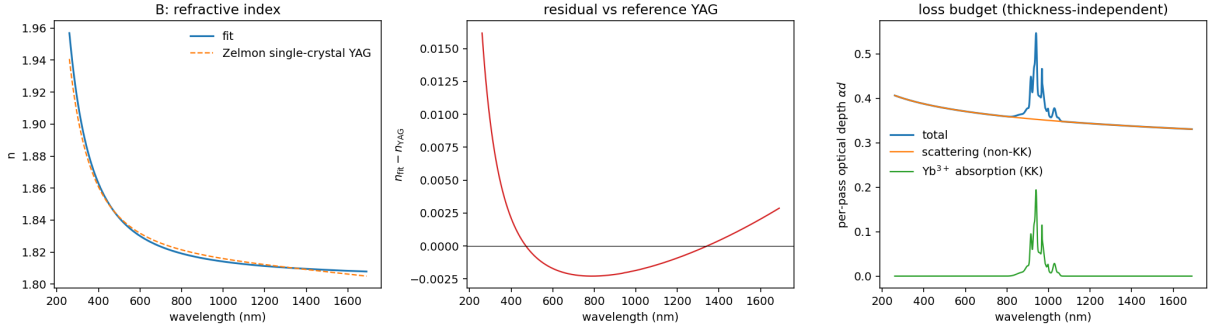


Figure 1: Sample B. **Left:** fitted $n(\lambda)$ against the single-crystal YAG reference. **Centre:** residual, within ± 0.0025 across 400–1600 nm. **Right:** the loss budget, separated into the non-KK scattering channel and the KK-consistent Yb³⁺ absorption. Scattering accounts for essentially all of the baseline loss; genuine absorption is confined to the Yb manifold.

8.2 Loss budget

$$\alpha_{\text{sc}}(\lambda) = (3.505 \pm 0.002) \left(\frac{1000 \text{ nm}}{\lambda} \right)^{0.1102 \pm 0.0010} \text{ cm}^{-1}. \quad (27)$$

The near-grey exponent $p = 0.110$ is the principal physical result. The dominant loss in these samples is neither absorption nor Rayleigh scattering: it is almost wavelength-independent, i.e. Mie/geometric scattering from features comparable to or larger than the wavelength — grain boundaries and residual pores. This is what the earlier analysis’s unexplained 2–3% NIR transmittance gap actually was. In per-pass optical depth, scattering falls from $\alpha d = 0.388$ at 400 nm to 0.335 at 1550 nm.

8.3 Yb³⁺ oscillators

All centres fixed at their predicted positions; only A and the width refine. Kind “e” denotes an electronic Stark transition, “v” a vibronic sideband.

Centre (nm)	Kind	A	FWHM (eV)	Assignment (cm^{-1})
875.1	v	$2.10 \times 10^{-6} \pm 3 \times 10^{-7}$	0.100*	10927 + 500
898.7	v	$3.46 \times 10^{-6} \pm 5 \times 10^{-7}$	0.0368	10927 + 200
915.2	e	$1.92 \times 10^{-5} \pm 9 \times 10^{-7}$	0.0140	$0 \rightarrow 10927$
930.5	v	$2.11 \times 10^{-5} \pm 8 \times 10^{-7}$	0.0162	ZPL + 420
940.4	e	$3.72 \times 10^{-5} \pm 1 \times 10^{-6}$	0.0118	$0 \rightarrow 10634$ (pump band)
954.5	v	$1.46 \times 10^{-5} \pm 4 \times 10^{-7}$	0.0831	ZPL + 150
965.1	e	$1.6 \times 10^{-6} \pm 1 \times 10^{-6}$	0.0020*	$565 \rightarrow 10927$
968.5	e	$2.09 \times 10^{-5} \pm 1 \times 10^{-6}$	0.0040	$0 \rightarrow 10327$ (ZPL) + $612 \rightarrow 10927$
993.1	e	$1.27 \times 10^{-6} \pm 1 \times 10^{-6}$	0.0026*	$565 \rightarrow 10634$
997.8	e	$3.62 \times 10^{-6} \pm 1 \times 10^{-6}$	0.0062	$612 \rightarrow 10634$
1024.4	e	$1.51 \times 10^{-6} \pm 2 \times 10^{-6}$	0.0191	$565 \rightarrow 10327$
1029.3	e	$6.89 \times 10^{-6} \pm 2 \times 10^{-6}$	0.0183	$612 \rightarrow 10327$
1048.0	e	$2.13 \times 10^{-6} \pm 9 \times 10^{-7}$	0.0151	$785 \rightarrow 10327$

* width at a bound; that parameter's uncertainty is not meaningful.

Not detected. 986.0 nm ($785 \rightarrow 10927$) and 1015.3 nm ($785 \rightarrow 10634$) refined to zero amplitude and were pruned. Both originate on the 785 cm^{-1} level, whose 300 K population is only 2%, so their weakness is expected rather than surprising.

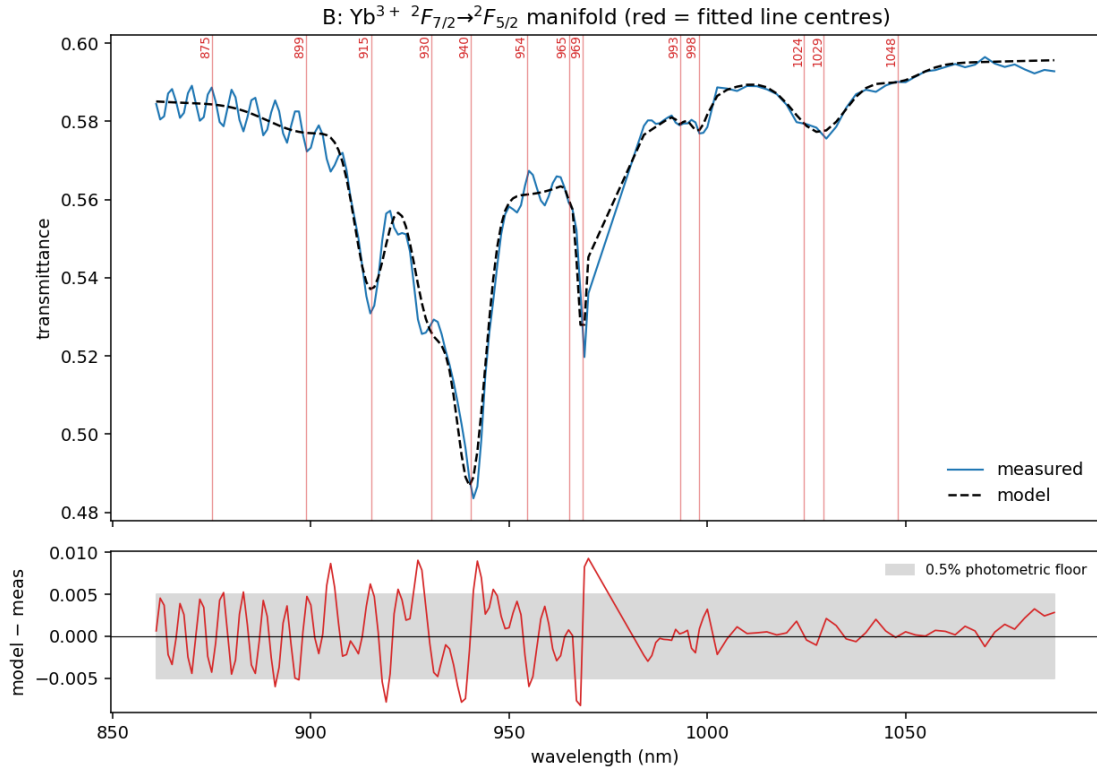


Figure 2: The $\text{Yb}^{3+} \ ^2F_{7/2} \rightarrow \ ^2F_{5/2}$ manifold. Red lines mark the *predicted* centres. The model reproduces the 915 nm line, the 921 nm peak between components, the 930 nm vibronic shoulder, the 941 nm pump band, the sharp 969 nm zero-phonon line and the 997/1029 nm hot bands. Lower panel: residual against the 0.5% photometric floor (grey); the regular oscillation below 950 nm is instrumental (Sec. 9.1), not material, and is deliberately not fitted.

8.4 Fit quality

Transmittance residual, 880–1080 nm	RMS	max
3 broad Gaussians (first attempt)	0.0087	0.0198
8 lines, centres free	0.0026	0.0060
13 lines, centres fixed at Stark values	0.0036	0.0093

The free-centre model scores marginally better but is *not physical*: its lines sit where the residual happens to be, not where Yb^{3+} has transitions. The fixed-centre model is the one to quote.

8.5 Cross-validation

Sample A was fitted from transmittance alone with n frozen at B’s reflection-derived dispersion. It reproduces B’s oscillator strengths to 0.1–0.9%:

Line	B ($R + T$)	A (T only)
915.2 nm	1.919×10^{-5}	1.920×10^{-5}
940.4 nm	3.722×10^{-5}	3.693×10^{-5}
954.5 nm (ZPL+150)	1.462×10^{-5}	1.449×10^{-5}
968.5 nm (ZPL)	2.092×10^{-5}	2.090×10^{-5}
Scattering C, p	3.505, 0.110	3.530, 0.131

Since the two samples were measured independently and A’s fit never sees reflection data, this is a genuine consistency check rather than a restatement of the same numbers.

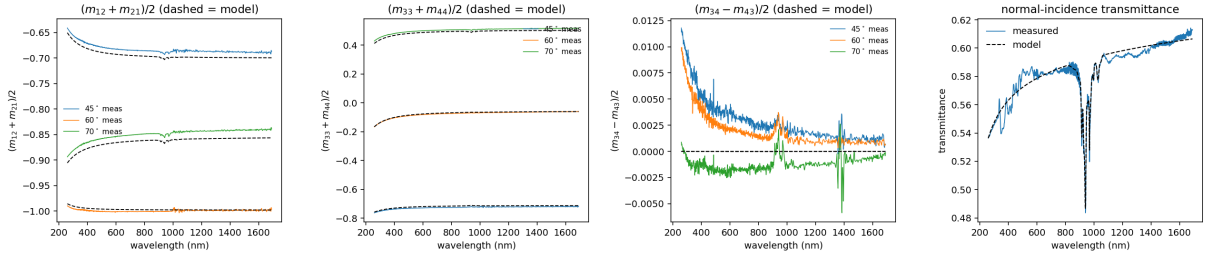


Figure 3: Measured (solid) and modelled (dashed) symmetrised Mueller elements at three angles of incidence, and the normal-incidence transmittance. The s_{34} panel shows the one systematic residual the isotropic model does not capture (Sec. 11).

9 Diagnostics: how the model was falsified and corrected

This section records the tests that changed the analysis. Each was prompted by a residual the model could not explain.

9.1 Which transmission dips are real

The measured transmittance contains **41 dips** with prominence > 0.004 . Fitting oscillators to all of them would have injected fictitious k and, through Eq. (12), corrupted n . They therefore had to be classified first.

The test. Reflection was acquired on 2026-06-30 and transmission on 2026-06-18. Genuine absorption raises k , which attenuates the backside beam and so *must* appear in the reflection Mueller matrix and in the depolarisation. An artefact confined to the transmission channel will not. Correlating the transmittance ripple against independently measured reflection observables:

Band	corr(T , depol)	corr(T , s_{12})	Verdict
300–500 nm	−0.09	+0.04	instrumental
500–700 nm	−0.06	+0.08	instrumental
700–900 nm	+0.05	−0.22	instrumental
900–1060 nm	+0.34	−0.42	real absorption
1100–1690 nm	+0.06	+0.20	instrumental

Only the Yb manifold is material. The remaining ~ 30 dips — at 349, 403, 467, 486, 544, 581, 602, 611, 756 nm, the dense 800–880 nm series, 1105 and 1527 nm — live purely in the transmission channel and are not fitted.

A tempting but invalid argument. Samples A and B reproduce this ripple at $r = 0.98$ – 0.99 . That is *not* evidence that it is real, because both share the same instrument. Only the reflection cross-check settles it.

Alternatives excluded. The 800–950 nm ripple is not a detector-interleave artefact: the sign alternation of consecutive differences is 41–44% (random is 50%) and the even/odd sample offset is 6×10^{-5} . Nor is it a single etalon: an FFT in wavenumber returns mutually inconsistent optical paths (151, 2.8, 15.9 μm) in different bands.

9.2 The detector mask was too wide

The Si/InGaAs crossover was initially masked over 965–983 nm. That range *swallowed the real zero-phonon line*. Inverting the transmittance at known n shows αd rising smoothly to 0.48 at 969 nm — physical absorption — whereas only at 971–983 nm does the transmittance jump to 0.60–0.68, above the sample’s own out-of-band baseline (≈ 0.58) and therefore unphysical.

With the ZPL masked out, the fit had no oscillator for the absorption climbing into the mask and compensated with the spurious 52 nm-wide “line” at 955 nm. Narrowing the mask to 971–983 nm and adding the ZPL+150 cm^{-1} vibronic replica removed the artefact and brought the 950–965 nm residual below the photometric floor.

9.3 A parameter-aliasing bug

The routine mapping parameter names to oscillator indices parsed `gauss12_amp` with `int(head[-1])` — the trailing *character*. With ten or more oscillators this maps `gauss10..gauss13` onto indices 0..3, silently aliasing four parameters onto four others.

Symptoms, all initially misread as physics: every uncertainty returned NaN (singular Jacobian); amplitudes were driven to absurd values (9.95×10^{-4} against a bound, versus $\sim 10^{-5}$ for real lines); and convergence took 697 function evaluations instead of 45. The printed per-line table looked plausible throughout, because it reads the model dictionary directly and never saw the aliased vector. The index is now parsed with a regular expression, and `python fit/dispersion.py` contains a regression test that round-trips 14 oscillators through pack/unpack.

The general lesson: a reporting path that bypasses the code under test will hide the bug it is meant to reveal.

10 Reproduction

```
python -m fit.run_fit --sample B --thickness 1.0    # full R(10 AOI) + T fit
python -m fit.run_fit --sample A                    # reuses B's dispersion
python -m fit.run_fit --list                         # incl. screened-out files
python -m fit.plot_fit --sample B                   # figures
python fit/dispersion.py                             # KK + pack/unpack self-tests
python fit/slab.py                                   # forward-model self-tests
```

Outputs land in `fit/out/`: `fit*.nk.csv` (per-wavelength n , k , α , αd , modelled and measured observables), `fit*.model.json` (parameters, uncertainties, line assignments) and the figures reproduced above.

11 Limitations and next steps

1. **Thickness.** Only αd is measured. Supplying a micrometer measurement converts every amplitude to an absolute cross-section.
2. **Residual anisotropy.** s_{34} shows a systematic rise towards the blue, ≈ 0.012 at 260 nm decaying to ≈ 0.002 in the NIR, which an isotropic model predicts as zero. Capturing it requires a uniaxial substrate; the implied Δn is $\sim 10^{-3}$.
3. **Two widths sit at bounds** (875.1 nm upper, 965.1 nm lower); their uncertainties are not meaningful.
4. **Samples 5 % and 10 % must be re-measured.** Their transmission files are corrupt and the 5 % reflection scans show no polarisation change at 30° – 50° , indicating a failed alignment.
5. **Absolute Yb concentration** is not determined here; the oscillator strengths are per unit thickness and would need a calibrated cross-section, or a known reference sample, to convert to ion density.

References

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